# Chapter 19: Inference about a Population Proportion

The Practice of Statistics in the Life Sciences
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Part IV: Specific Inference Procedures

# Why Chapter 19 Matters

Earlier chapters developed inference for population means using quantitative data.

Many scientific questions instead record a categorical outcome with two possible categories.

- Does a person currently test positive for an infection?
- Does a patient respond to treatment?
- Does an animal display a particular behaviour?

**Main Question**

How can a sample proportion be used to estimate or test an unknown population proportion?

# Chapter Roadmap

The chapter develops five connected ideas.

1. The sample proportion and its sampling distribution
2. Large-sample confidence intervals
3. The more accurate plus four confidence interval
4. Sample-size planning for a desired margin of error
5. One-proportion hypothesis tests

The theory comes first. Each procedure is then applied to a life-science example and reproduced in R.

# Means and Proportions

| | **Quantitative response** | **Binary categorical response** |
|---|---|---|
| Population parameter | Mean $\mu$ | Proportion $p$ |
| Sample statistic | Mean $\bar{x}$ | Proportion $\hat{p}$ |
| Typical question | Average measurement | Fraction with an outcome |
| Reference model | $t$ or Normal | Approximately Normal |

The inferential logic remains the same: use a sample statistic to learn about an unknown population parameter while accounting for sampling variation.

# Defining the Outcome

For convenience, the outcome being counted is called a **success**. The label does not imply that the outcome is desirable.

Before collecting or analyzing data, define

- The target population
- The observational unit
- The exact rule that identifies a success
- The time period or measurement window

## Why the Definition Matters

Different definitions of *smoker*, *infection* or *treatment response* can produce very different proportions from the same population.

# The Sample Proportion

Suppose a sample contains $n$ individuals and $X$ successes.

## Sample Proportion

$$\hat{p} = \frac{X}{n}$$

The statistic $\hat{p}$ estimates the unknown population proportion $p$.

- $\hat{p}$ is read as *p-hat*
- $0 \leq \hat{p} \leq 1$
- $100\hat{p}\%$ is the corresponding sample percentage
- A different random sample usually produces a different value of $\hat{p}$

# Example 19.1: HPV Prevalence

A nationally representative survey obtained valid specimens from 1921 women aged 14 to 59. Of these specimens, 515 tested positive for HPV.

## Sample Estimate

$$\hat{p} = \frac{515}{1921} \approx 0.2681$$

$$100\hat{p}\% \approx 26.81\%$$

The observed percentage describes the sample. Inference is needed to estimate the proportion $p$ of all women in the target population who would test positive.

# Sampling Distribution of $\hat{p}$

For a random sample of size $n$ from a large population with proportion $p$,

**Centre**

$$\mu_{\hat{p}} = p$$

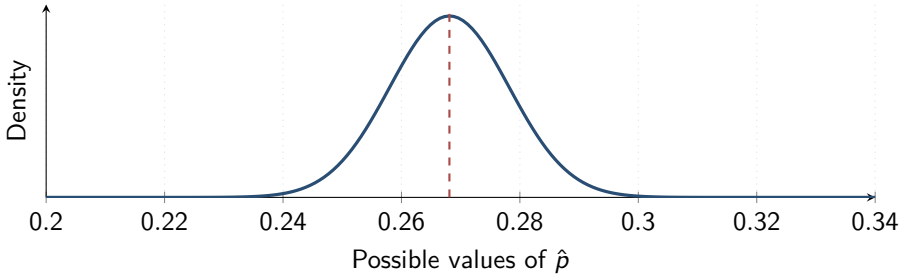
**Spread**

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

The sample proportion is an unbiased estimator of $p$. Its sampling variability decreases as the sample size increases.

# Shape of the Sampling Distribution

When the expected numbers of successes and failures are sufficiently large, the sampling distribution of $\hat{p}$ is approximately Normal.




The approximation can be poor when $n$ is small or when $p$ is close to 0 or 1.

# Conditions for Inference about a Proportion

Inference about one population proportion requires three ideas.

- **Randomness:** The data can be treated as a simple random sample or arise from another defensible random process
- **Independence:** The population is much larger than the sample, often checked with $n \leq 0.10N$
- **Large counts:** The relevant procedure has enough expected or observed successes and failures for its Normal approximation

## Design Before Formula

A larger sample reduces random sampling error. It does not remove selection bias, nonresponse bias or an unclear definition of success.

# Anatomy of a Confidence Interval

A confidence interval for a population proportion has the familiar form

$$\text{estimate} \pm \text{critical value} \times \text{standard error}$$

Using $\hat{p}$ to estimate $p$ gives

$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

The margin of error is

$$m = z^* SE_{\hat{p}}$$

The critical value $z^*$ comes from the Standard Normal distribution.

# Large-Sample Confidence Interval

For a random sample from a large population, the approximate level $C$ interval is

## Large-Sample Interval

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Common critical values are

| Confidence level | $z^*$ |
|---|---|
| 90% | 1.645 |
| 95% | 1.960 |
| 99% | 2.576 |

# Why the Critical Value Is $z^*$

Inference for a mean uses a $t$ distribution when the separate spread parameter $\sigma$ must be estimated by $s$.

For a proportion, the sampling variance is determined by the same parameter being estimated:

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

Replacing $p$ with $\hat{p}$ does not create a $t$ statistic. The method continues to use a Normal approximation and a $z^*$ critical value.

# Conditions for the Large-Sample Interval

Use the ordinary large-sample interval only when

- The sampling design supports inference to the target population
- The population is much larger than the sample
- The sample contains at least 15 successes
- The sample contains at least 15 failures

$$X \geq 15 \qquad \text{and} \qquad n - X \geq 15$$

Observed counts are used because the population proportion $p$ is unknown.

# Example 19.3: HPV Confidence Interval

## State

Estimate the proportion $p$ of women aged 14 to 59 in the target population who currently test positive for HPV.

## Plan

Construct a 95% large-sample confidence interval using $X = 515$ positive specimens from $n = 1921$ valid specimens.

The point estimate is

$$\hat{p} = \frac{515}{1921} = 0.2681$$

# Example 19.3: Checking the Conditions

- NHANES used a complex representative sampling design. The analysis treats its overall behaviour as approximately comparable to an SRS
- The target population is much larger than 1921
- The number of successes is 515
- The number of failures is $1921 - 515 = 1406$

$$515 \geq 15 \qquad \text{and} \qquad 1406 \geq 15$$

The count condition is easily satisfied. The sampling design requires more careful interpretation than the calculation alone suggests.

# Example 19.3: Calculating the Interval

The estimated standard error is

$$SE = \sqrt{\frac{(0.2681)(0.7319)}{1921}} \approx 0.01011$$

The margin of error is

$$m = 1.960(0.01011) \approx 0.01981$$

The 95% confidence interval is

$$0.2681 \pm 0.01981$$

$$\boxed{(0.2483,\ 0.2879)}$$

# Example 19.3: Calculation in R

```r
# Observed count and sample size
x_hpv <- 515
n_hpv <- 1921

# Large-sample 95% confidence interval
p_hat_hpv <- x_hpv / n_hpv
se_hpv <- sqrt(p_hat_hpv * (1 - p_hat_hpv) / n_hpv)
z_star <- qnorm(0.975)
margin_hpv <- z_star * se_hpv

c(
  estimate = p_hat_hpv,
  lower = p_hat_hpv - margin_hpv,
  upper = p_hat_hpv + margin_hpv
)
```

# Example 19.3: Interpretation and Scope

We are 95% confident that the proportion of women aged 14 to 59 in the target population who would currently test positive for HPV lies between approximately 0.248 and 0.288.

Equivalently, the estimated population percentage is between approximately 24.8% and 28.8%.

## Study Limitation

About 23% of the original sample did not provide a usable specimen. If respondents and nonrespondents differ in HPV prevalence, nonresponse can bias the estimate. The margin of error does not measure that bias.

# Limitations of the Ordinary Interval

The large-sample interval is easy to calculate, but its actual coverage can fall below the requested confidence level.

Problems are most noticeable when

- The sample size is modest
- The number of successes is small
- The number of failures is small
- The true population proportion is close to 0 or 1

A simple adjustment improves coverage substantially in many of these settings.

# The Plus Four Method

The plus four method adds four imaginary observations:

- Two successes
- Two failures

The adjusted estimate is

**Plus Four Estimate**

$$\widetilde{p} = \frac{X + 2}{n + 4}$$

The confidence interval then uses the ordinary formula with $\widetilde{p}$ and the adjusted sample size $n + 4$.

# Plus Four Confidence Interval

The plus four interval is

$$\widetilde{p} \pm z^* \sqrt{\frac{\widetilde{p}(1-\widetilde{p})}{n+4}}$$

Use this method when

- The confidence level is at least 90%
- The original sample size satisfies $n \geq 10$
- The sampling and independence conditions are reasonable

The adjustment moves the estimate slightly toward 0.5, where the Normal approximation behaves more reliably.

# Example 19.4: Blinding in Clinical Trials

Investigators randomly selected 97 articles that reported placebo-controlled clinical trials. Only 7 articles discussed whether blinding had succeeded.

**Question**

What proportion $p$ of all similar articles discuss the success of blinding?

The ordinary large-sample interval is not recommended because the sample contains only 7 successes. The sample size is large enough for a plus four interval.

# Example 19.4: Plus Four Calculation

The adjusted estimate is

$$\widetilde{p} = \frac{7+2}{97+4} = \frac{9}{101} \approx 0.0891$$

The adjusted standard error is

$$SE = \sqrt{\frac{(0.0891)(0.9109)}{101}} \approx 0.02835$$

The 95% plus four interval is

$$0.0891 \pm 1.960(0.02835) = \boxed{(0.0335,\ 0.1447)}$$

# Example 19.4: Plus Four Interval in R

```r
# Original study results
x_blind <- 7
n_blind <- 97

# Add two successes and two failures
x_plus4 <- x_blind + 2
n_plus4 <- n_blind + 4
p_plus4 <- x_plus4 / n_plus4

se_plus4 <- sqrt(p_plus4 * (1 - p_plus4) / n_plus4)
margin_plus4 <- qnorm(0.975) * se_plus4

c(
  estimate = p_plus4,
  lower = p_plus4 - margin_plus4,
  upper = p_plus4 + margin_plus4
)
```

# Ordinary and Plus Four Intervals

For the blinding example,

| Method | Estimate | Lower bound | Upper bound |
|---|---|---|---|
| Ordinary large-sample | 0.0722 | 0.0207 | 0.1237 |
| Plus four | 0.0891 | 0.0335 | 0.1447 |

The numerical difference is modest, but the repeated-sampling coverage of the plus four method is more reliable.

## Recommended Method

Use the plus four interval when estimating one population proportion under its stated conditions.

# Choosing a Sample Size

To plan for a margin of error no greater than $m$, use

## Required Sample Size

$$n = \left(\frac{z^*}{m}\right)^2 p^*(1 - p^*)$$

Here, $p^*$ is a planning value for the anticipated population proportion.

Always round the calculated sample size **up**. Rounding down can produce a margin of error larger than requested.

# Choosing the Planning Value $p^*$

Two approaches are common.

- Use a value from a pilot study, previous research or credible subject-matter knowledge
- Use $p^* = 0.5$ when no useful prior estimate is available

The product $p^*(1 - p^*)$ is largest at $p^* = 0.5$:

$$(0.5)(1 - 0.5) = 0.25$$

This choice gives the largest required sample size and therefore provides a conservative plan.

# Example 19.5: Planning a Survey

A university wants to estimate the proportion of students who never floss. The desired confidence level is 95%, and the margin of error should be no greater than 0.04.

With no precise prior estimate, use $p^* = 0.5$:

$$n = \left(\frac{1.960}{0.04}\right)^2 (0.5)(0.5) = 600.25$$

Round upward:

$$\boxed{n = 601}$$

# Sample-Size Planning in R

```r
# Desired design values
confidence <- 0.95
margin <- 0.04
p_star <- 0.50

# Critical value and required sample size
z_star <- qnorm(1 - (1 - confidence) / 2)
n_required <- (z_star / margin)^2 *
  p_star * (1 - p_star)

ceiling(n_required)
# [1] 601
```

For margins of error 0.03 and 0.02, the required sample sizes are 1068 and 2401, respectively.

# Hypothesis Tests for a Proportion

A one-proportion test evaluates a specific claim about the population proportion.

$$H_0 : p = p_0$$

Under the null hypothesis, the sampling distribution of $\hat{p}$ has centre $p_0$ and standard deviation

$$\sqrt{\frac{p_0(1-p_0)}{n}}$$

The null value $p_0$, rather than the observed $\hat{p}$, determines the standard error used in the test statistic.

# The One-Proportion $z$ Statistic

The test statistic is

## Test Statistic

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

Use the Normal approximation when the expected counts under $H_0$ satisfy

$$np_0 \geq 10 \qquad \text{and} \qquad n(1 - p_0) \geq 10$$

The test uses expected counts under the null hypothesis. Confidence-interval conditions use observed counts.

# Alternative Hypotheses and Tail Areas

The alternative hypothesis determines the P-value.

| Alternative | Direction | P-value |
|---|---|---|
| $H_a : p > p_0$ | Upper tail | $P(Z \geq z)$ |
| $H_a : p < p_0$ | Lower tail | $P(Z \leq z)$ |
| $H_a : p \neq p_0$ | Two tails | $2P(Z \geq \lvert z \rvert)$ |

Choose the direction from the research question before examining the sample result.

# Example 19.6: Aphid Righting Behaviour

Researchers released 20 live pea aphids upside down. Nineteen landed on their legs.

## Research Question

Do live aphids land on their legs more often than the probability 0.5 expected from chance alone?

Let $p$ be the population proportion that land on their legs.

$$H_0 : p = 0.5 \qquad H_a : p > 0.5$$

Under $H_0$, both expected counts equal $20(0.5) = 10$, so the count condition is satisfied.

# Example 19.6: Test Statistic and P-Value

The sample proportion is

$$\hat{p} = \frac{19}{20} = 0.95$$

The null standard error is

$$SE_0 = \sqrt{\frac{(0.5)(0.5)}{20}} \approx 0.1118$$

The test statistic is

$$z = \frac{0.95 - 0.5}{0.1118} \approx 4.025$$

The upper-tail P-value is

$$P(Z \geq 4.025) \approx 0.0000285$$

# Example 19.6: Test in R

```r
# Observed data and null value
x_aphid <- 19
n_aphid <- 20
p0 <- 0.50

# One-proportion z test calculated directly
p_hat_aphid <- x_aphid / n_aphid
se_null <- sqrt(p0 * (1 - p0) / n_aphid)
z_aphid <- (p_hat_aphid - p0) / se_null

# Upper-tail P-value for Ha: p > 0.50
p_value <- 1 - pnorm(z_aphid)

c(z = z_aphid, p_value = p_value)
```

# Example 19.6: Conclusion and Limitations

The very small P-value provides strong evidence against $H_0$.

The data support the conclusion that live pea aphids land on their legs more often than chance alone would predict under the study conditions.

## What the Test Does Not Establish

The test does not identify the biological mechanism responsible for the behaviour. It also does not quantify how well the laboratory aphids represent every population or natural dropping condition.

Statistical significance addresses compatibility with the null model. Scientific interpretation still depends on the design and biological context.

# Choosing the Correct Procedure

| **Research goal** | **Procedure** |
|---|---|
| Estimate one population proportion with large observed counts | Large-sample interval |
| Estimate one population proportion with $n \geq 10$ and confidence at least 90% | Plus four interval |
| Plan a sample for a target margin of error | Sample-size formula using $p^*$ |
| Test $H_0 : p = p_0$ with adequate expected counts | One-proportion $z$ test |
| Compare proportions from two groups | Two-proportion methods in Chapter 20 |

The study design and definition of success determine whether the calculation answers the intended question.

# Chapter 19 Summary

- The sample proportion $\hat{p} = X/n$ estimates the population proportion $p$
- The sampling distribution has centre $p$ and standard deviation $\sqrt{p(1-p)/n}$
- The ordinary interval uses $\hat{p} \pm z^* \sqrt{\hat{p}(1-\hat{p})/n}$
- The plus four interval usually provides more accurate coverage
- Sample-size planning uses $n = (z^*/m)^2 p^*(1-p^*)$
- A test of $H_0 : p = p_0$ uses the null standard error based on $p_0$

## Final Principle

The procedure measures random sampling uncertainty. Population definitions, sampling quality and study conditions determine the scope of the conclusion.